Wireless Network Using Cisco Packet Tracer

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1. Introduction

The fast growth of digital communication systems has forced organisations to come up with network infrastructures which are not only scalable and efficient, but also that are able to support heterogeneous devices and technologies. The modern networks normally combine both the wired and wireless means of communication in order to improve flexibility, mobility and performance. In this coursework, a full hybrid network design was developed, configured and simulated with Cisco Packet Tracer in a systematic approach (Baek et al., 2021). It was a project that included the calculation and implementation of individual student identification number-based block IP address, breaking down the address into several logical networks, setting up routing and switching features, supporting secure wireless internet connectivity, and ensuring a complete end-to-end communication of all devices (Ashraf et al., 2023). The report gives a clear description of conceptualisation, design, logical planning, configuration, and testing of the real implementation of the network.

The major objective of the design was to establish a working enterprise type network, which involves wired segments, wireless gadgets, routers, switches, servers and security systems. Through a modular and methodologically directed design, the network was built on industry standard settings, i.e. VLAN segmentation, router-on-a-stick inter-VLAN routing, DHCP, dynamic routing protocols and wireless access control (Hadiwinata et al., 2025). All these factors lead to the reliability, scalability and security of the resulting network in general.

2. Network Requirements and Design Overview

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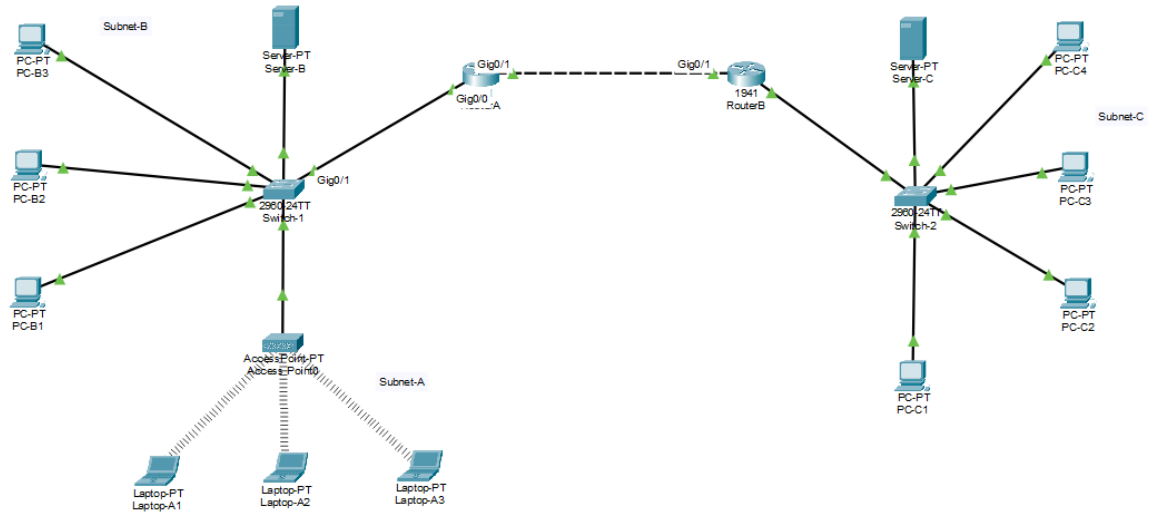

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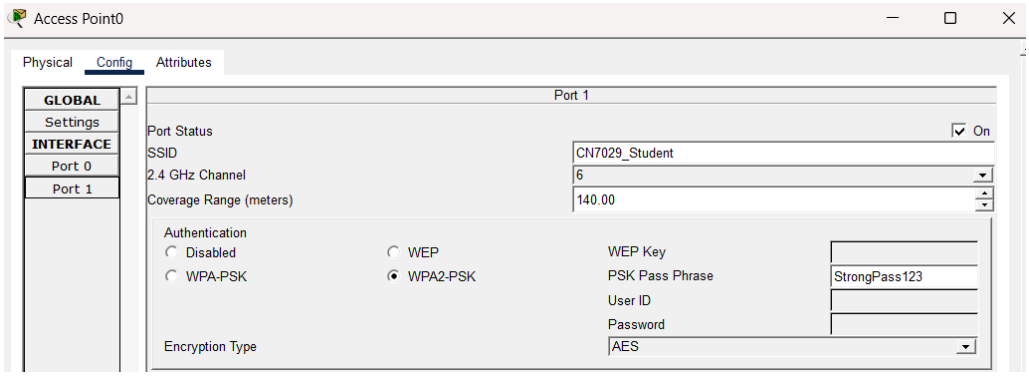
Subnet	Purpose	Network Address	CIDR	Mask
A	Wireless (VLAN 10)	191.52.31.0	/27	255.255.255.224
B	Wired (VLAN 20)	191.52.31.32	/27	255.255.255.224
C	Wired (Subnet C)	191.52.31.64	/27	255.255.255.224
P2P	Router-A–Router-B link	191.52.31.96	/30	255.255.255.252


Access Point0
Physical
Config
Attributes
GLOBAL
Settings
INTERFACE
Port 0
Port 1
Port 1
Port Status
On
SSID
CN7029_Student
2.4 GHz Channel
6
Coverage Range (meters)
140.00
Authentication
Disabled
WEP
WPA-PSK
WPA2-PSK
WEP Key
PSK Pass Phrase
StrongPass123
User ID
Password
Encryption Type
AES

Switch-1

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch(config)#vlan 10
Switch(config-vlan)#name Wireless
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name WiredB
Switch(config-vlan)#exit
Switch(config)#interface range fa0/2 - fa0/4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interface fa0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface gigabitEthernet0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

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RouterA

Physical Config CLI Attributes

IOS Command Line Interface

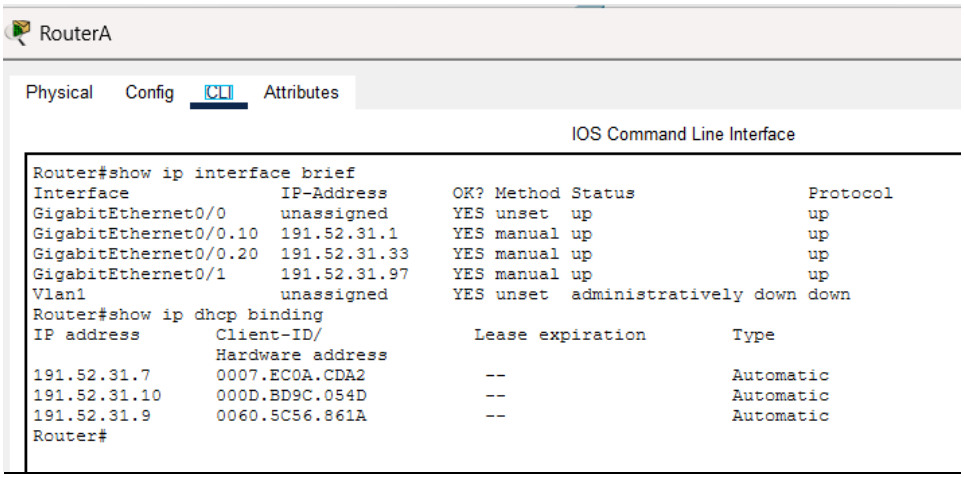
```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 191.52.31.1 255.255.255.224
Router(config-subif)#no sutdown
                      ^
% Invalid input detected at '^' marker.

Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface GigabitEthernet0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 191.52.31.33 255.255.255.224
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

[redacted]



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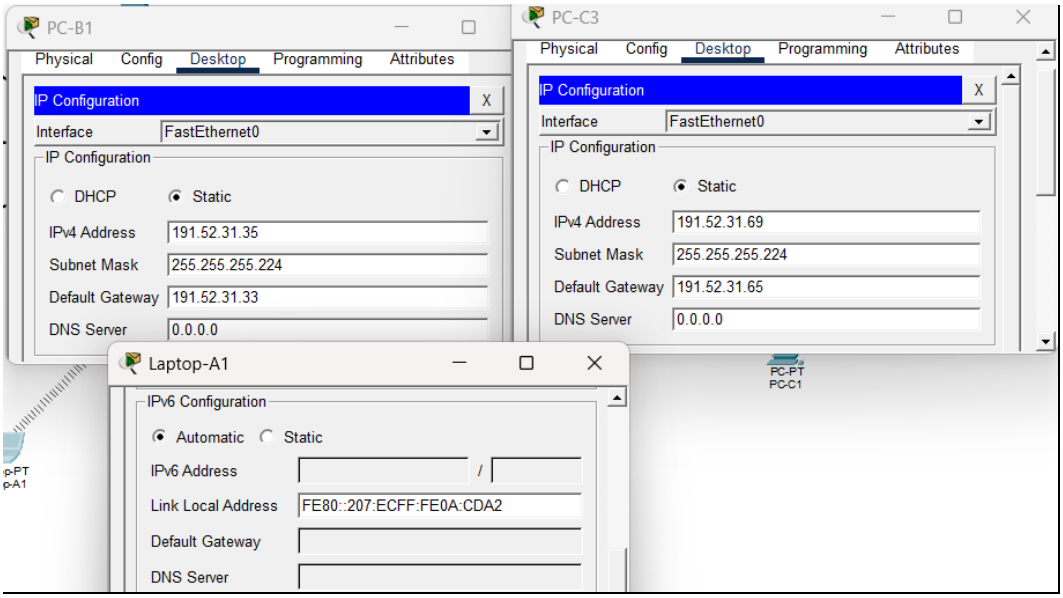

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RouterA

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#exit
Router(config)#ip dhcp excluded-address 191.52.31.1 191.52.31.5
Router(config)#ip dhcp pool VLAN10_POOL
Router(dhcp-config)#network 191.52.31.0 255.255.255.224
Router(dhcp-config)#default-router 191.52.31.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 191.52.31.33 191.52.31.40
Router(config)#ip dhcp pool VLAN20_POOL
Router(dhcp-config)#network 191.52.31.32 255.255.255.224
Router(dhcp-config)#default-router 191.52.31.33
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
```

[redacted]

RouterB

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 191.52.31.98 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 191.52.31.65 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#ip dhcp excluded-address 191.52.31.65 191.52.31.68
Router(config)#ip dhcp pool SUBNETC_POOL
Router(dhcp-config)#network 191.52.31.64 255.255.255.224
Router(dhcp-config)#default-router 191.52.31.65
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
```

[redacted]

[redacted]

RouterA

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 191.52.31.0
Router(config-router)#network 191.52.31.96
Router(config-router)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
```

[redacted]

RouterB

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 191.52.31.64
Router(config-router)#network 191.52.31.96
Router(config-router)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
```

[redacted]

RouterA

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
191.52.31.7      0007.EC0A.CDA2           --                     Automatic
191.52.31.10     000D.BD9C.054D           --                     Automatic
191.52.31.9      0060.5C56.861A           --                     Automatic
Router#
```

[redacted]

PC-B1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 191.52.31.36

Pinging 191.52.31.36 with 32 bytes of data:

Reply from 191.52.31.36: bytes=32 time=1ms TTL=128
Reply from 191.52.31.36: bytes=32 time<1ms TTL=128
Reply from 191.52.31.36: bytes=32 time<1ms TTL=128
Reply from 191.52.31.36: bytes=32 time=1ms TTL=128

Ping statistics for 191.52.31.36:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

[redacted]

Laptop-A1

```
C:\>
ping 191.52.31.1

Pinging 191.52.31.1 with 32 bytes of data:

Reply from 191.52.31.1: bytes=32 time=43ms TTL=255
Reply from 191.52.31.1: bytes=32 time=23ms TTL=255
Reply from 191.52.31.1: bytes=32 time=11ms TTL=255
Reply from 191.52.31.1: bytes=32 time=9ms TTL=255

Ping statistics for 191.52.31.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 9ms, Maximum = 43ms, Average = 21ms

C:\>ping 191.52.31.1

Pinging 191.52.31.1 with 32 bytes of data:

Reply from 191.52.31.1: bytes=32 time=24ms TTL=255
Reply from 191.52.31.1: bytes=32 time=26ms TTL=255
Reply from 191.52.31.1: bytes=32 time=24ms TTL=255
Reply from 191.52.31.1: bytes=32 time=24ms TTL=255
```

Top

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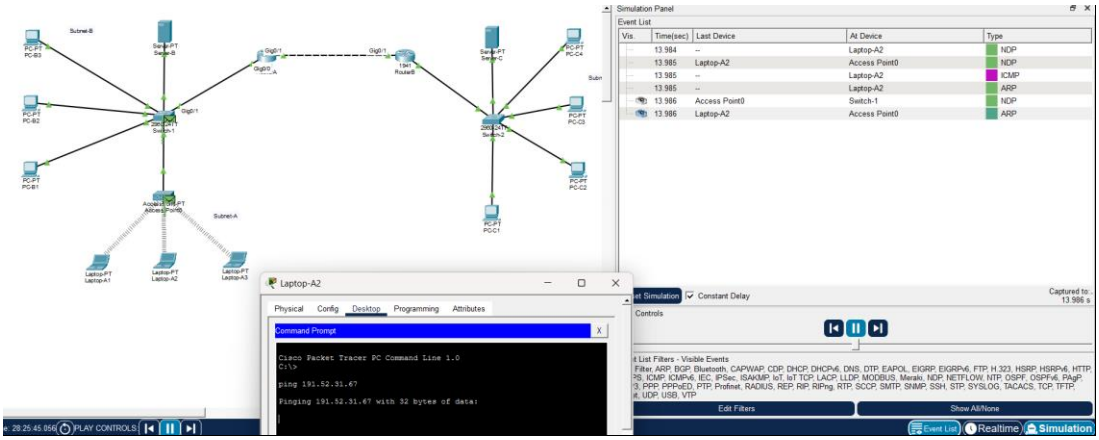

Laptop-A2

```
Pinging 191.52.31.67 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 191.52.31.67: bytes=32 time=19ms TTL=126
Reply from 191.52.31.67: bytes=32 time=11ms TTL=126

Ping statistics for 191.52.31.67:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 19ms, Average = 15ms

C:\>ping 191.52.31.67

Pinging 191.52.31.67 with 32 bytes of data:

Reply from 191.52.31.67: bytes=32 time=27ms TTL=126
Reply from 191.52.31.67: bytes=32 time=6ms TTL=126
Reply from 191.52.31.67: bytes=32 time=12ms TTL=126
Reply from 191.52.31.67: bytes=32 time=17ms TTL=126

Ping statistics for 191.52.31.67:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
```

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]